

Research article

Detecting management gaps for biodiversity conservation: An integrated assessment

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ABSTRACT

The latest report on the state of nature in Europe (2013–2018) shows that biodiversity is declining at an alarming rate, with most protected species and habitats in poor condition. Despite an increasing volume of collected biodiversity information, urgent action is needed to integrate biodiversity data and knowledge to improve conservation efforts.

We conducted a study in Catalonia (NE Spain), where we collected management measures implemented between 2013 and 2018, including allocation, budget, pressures aimed, and habitats/species potentially benefiting. We integrated information on pressures and habitats/species with the measures to identify non-spatial management gaps. Then, we integrated the spatially explicit information to determine the spatial management gap, identifying geographical areas where species/habitats are under pressure without registered measures. We demonstrated the importance of integrating existing information.

Our findings revealed that resources were often not distributed adequately across species/habitats, with biases towards certain taxa being a common issue. The non-spatial management gap analysis identified taxonomic groups, especially plants and mollusks with the wider management gaps. We also identified threatened areas, especially in the northeast of the region with the larger spatial management gaps. These results could guide priority objectives to optimize conservation efforts.

Integrating different information sources provided a broader view of the challenges that conservation science is facing nowadays. Our study offers a path toward bending the curve of biodiversity loss by providing an integrative framework that could optimize the use of the available information and help narrow the knowing-doing gap. In the context of the EU, this example demonstrates how information can be used to promote some environmental policy instruments, such as the Prioritized Action Frameworks (PAFs). Additionally, our findings highlight the importance of supporting decision-making with systematic assessments to identify deficiencies in the conservation process, reduce the loss of critical ecosystems and species, and avoid biases among taxa.

1. Introduction

Under the multiple evidence of the global decline of biodiversity and the ecosystem services it provides (S. Díaz et al., 2020; WWF et al.,

2022), there is agreement on the need for investment to reverse the situation (Ceballos et al., 2015; Convention on Biological Diversity, 2020). The international community, through regular conventions, has adopted a range of agreements aimed at addressing biodiversity loss.

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These conventions are supported by operational instruments that include the development of goals, targets, and work plans. The recent United Nations Biodiversity Conference (CBD COP15) held in December 2022 delineated specific targets to restore 30% of the degraded earth's surface, reduce the loss of areas of high biodiversity importance, and achieve effective conservation and management of 30% of the world's natural areas. The number of available assessments has grown in the previous two decades, which has been a cornerstone of biodiversity science to picture the complexity of the situation and achieve all these targets. For example, the IUCN Red List of Threatened Species and the Living Planet Index measure periodically the conservation status of thousands of species worldwide (IUCN, 2023; WWF et al., 2020). The volume of available data describing biodiversity distribution, functions, environmental variables, and ecosystem characteristics produced has grown exponentially due to advances in digital technologies (Runting et al., 2020). Substantial progress has been made in digitalizing and collecting information through the creation of data repositories, including historical data (Clavero et al., 2022). At a global scale, the Intergovernmental Science-policy Platform on Biodiversity and Ecosystem Services (IPBES), was created in 2013 to collect and mainstream the already existing information to coordinate and better respond to this global crisis (S. Díaz et al., 2019). The apparition of other data collection initiatives such as the Global Biodiversity Information Facility (GBIF), or the Global Database of Plants Traits (TRY) (Kattge et al., 2011) and also citizen science initiatives (Ellwood et al., 2017) enabled the scientific community to search for macroecology patterns and mechanisms at a global scale of species distributions, conservation assessments, or animal diseases (Heberling et al., 2021).

However, as the volume of information increases, so does the demand for tools to integrate data and biodiversity knowledge (König et al., 2019). Addressing policy demands in the context of the biodiversity crisis requires the integration of heterogeneous data sources and enhances improving transdisciplinary analyses (Wüest et al., 2020). Occupancy models, for example, are dealing with heterogeneous information challenges in species distribution modeling (Bailey et al., 2014). There are also studies integrating land use changes and climate change to address biodiversity losses (López, 2022; Rigal et al., 2023), or exploring the applicability of machine learning, remote sensing, and citizen science simultaneously for biodiversity conservation (Antonelli et al., 2022). In the face of this expanding pool of available and generated data, several studies call for a balance between data and/or knowledge creation and action, i.e. the knowing-doing gap (Groff et al., 2023; Habel et al., 2013). While many studies primarily describe the situation, only a few offered actual design or execution choices (La Salle et al., 2016; Williams et al., 2020). Addressing crucial questions such as the one we presented here about integrating information to assess where are we focusing our efforts and aligning them with areas where it is more needed, is essential to actively contribute to the needed changes to avoid biodiversity and ecosystem collapse (Fazey et al., 2018; Pereira et al., 2020).

At the European Union (EU) level, the Natura 2000 network, which resulted from the implementation of the Birds and Habitats Directives (EC, 2018) is the main tool to address biodiversity loss in Europe. Under this policy framework, there has been an increase in the EU's land surface designated as protected areas during the last decades. In 2021 protected areas covered 26% of the EU's land, including both Natura 2000 and national designations (EEA, 2020). In the same context, all Member States are bound to provide information every six years regarding the status of the populations of bird species included in Annex I of the Birds Directive and habitats and non-bird species included in Annexes I and II of the Habitats Directive. The last Report on EU Directives was developed for the period 2013 to 2018 (EEA, 2020). The main information compiled in this report includes 1) species distribution, trends and population size, and habitats distribution and surface; 2) pressures that are threatening species and habitats; and 3) measures that Member States are implementing to address the pressures identified. The

information gathered is used to develop an overall assessment of the conservation status of those species and habitats at the EU level. This assessment is mandatory for species/habitats of the Habitat Directives, no final assessment is requested for the Birds Directives. According to the most recent assessment (2013–2018), biodiversity is still declining across Europe, mainly as a consequence of pressures related to land use changes, modification of water regimes, pollution, invasive alien species, and climate change (EEA, 2020). Also, this reporting of conservation efforts is linked with other European policies. According to the art. 8 of the Habitats Directive, the European Commission shall adopt Prioritized Action Frameworks (PAFs) that define measures that should be adopted inside Natura 2000 and recently, outside protected areas as well (EC, 2022). These include measures in the green infrastructure network or even the ones designed through the new Common Agricultural Policy (Pe'er et al., 2020).

The use of integrated knowledge would guide priority conservation needs, identify gaps in conservation efforts, and support coordination between different conservation actors and stakeholders (Buxton et al., 2021). This is essential to address EU policy requests and to face the biodiversity crisis globally. Biodiversity assessments are proven to be key tools for monitoring ecosystems and species over time, although, turning this evidence into effective policies and actions is still a big challenge (Bradshaw, 2020). In practice, there is a lack of connection between monitoring conservation efforts and the planning and implementation of management measures (Rose et al., 2019). Conservation has not been a political priority, mismatches of timescales, and lack of funding are, among others, the causes of the lack of connection between scientific evidence and application (Rose et al., 2018). At the same time, conservation action efforts are often not optimized and not efficiently spatially distributed (Hermoso et al., 2019, 2022). In addition, conservation action effectiveness is not always monitored (Bennett et al., 2018). Another limitation is the unsuccessful measurement of the impact of measures, which difficult the application of evidence-based conservation. This is caused by the deficit of human and/or material resources, the lack of policy mechanisms or ineffective ones, or the disconnection between monitoring efforts and conservation plans (Moore et al., 2021; Waldron et al., 2013). Better informed conservation action is still required, specially adapted to the scale in which management decisions are made. In recent years, the increasing volume of available information has provided insight into pressures affecting biodiversity (Prakash and Verma, 2022). Understanding the specific locations of these pressures and their impacts on ecosystems and species is crucial to halt the biodiversity crisis. However, a notable gap exists concerning the alignment between where pressures occur and our conservation management strategies. Current literature emphasizes the importance of assessing the effectiveness of conservation measures (Sutherland et al., 2021) and protected areas, but there is limited knowledge regarding whether our conservation efforts are been taken in locations where they are most required. There is a need for increased efforts to systematically compile information on conservation measures (Mispiratcegy et al., 2021) and integrate this information with pressures occurring in the ecosystems.

Therefore, data integration and assessments are critical to 1) deal with policy requests and manage conservation measures to move from evidence to action, and 2) identify management gaps and conservation priorities in the conservation process. We understand as management gaps to sites or species facing pressures to their conservation, yet there are no identified management measures or ongoing efforts in place to address these challenges (Hermoso et al., 2022). We proposed a case study in Catalonia (NE Spain) to evaluate how multiple available data sources on pressures, measures, and species and habitats can be integrated to guide decision-making for biodiversity conservation. In particular, we collected information on the conservation measures for species or habitats cited in the last Report on EU Directives (period 2013–2018) with information about the spatial distribution of pressures and the sensitivity of each species/habitat to each pressure. We then

identified management gaps, i.e. pressures not covered by conservation measures aimed to reduce them. We assessed it both in a non-spatial context, evaluating the information integrated by groups of species/habitat, and spatially, converting data (species/habitat, pressures, and measures distributions) to a 1 km grid cell resolution across the study area. We wanted to show the wide potential applications of interconnected and organized information, and how the integration of different databases of already existing information in the territory in a structured and functional way could help identify management gaps and give recommendations on biodiversity conservation decision-making.

2. Methods

1. Study area

Catalonia (Spain) is in the northeast of the Iberian Peninsula, covering an area of 32,000 km² approx. It is a populated territory, with 7.5 million inhabitants and an average population density of 239 inhabitants per km². The region is located in the Mediterranean and Alpine bioclimatic regions, with a predominantly temperate Mediterranean climate and variations influenced by orographic: colder weather in the Pyrenees mountains and warmer temperatures on the coastline. Forest and natural areas are dominant in the region, implying 60 % of the surface, followed by different intensities of agricultural land (25–30 %). It is a territory with high biodiversity values due to its heterogeneous landscapes including 245 species and 99 habitats included in the Birds and Habitats EU Directives annexes, which implied that are declared of Community interest by the UE, including some endemism (Brotons et al., 2020). As well, 30% of the whole territory is declared under the Natura 2000 network, these protected areas had traditionally coexisted with human uses (Fig. 1).

2. Data collection

We collected information on conservation measures implemented in Catalonia during the time of the last Report on EU Directives (2013–2018). The collection of information was mandatory for every Member State to achieve a report on the species and habitats included in Annexes I and II of the Habitats Directive and Annex I of the Birds Directive. The temporal window used was decided based on the European policy demands for this report every six years, and it is common for

all Member States. We integrated these conservation measures information with different pieces of information to show the potential value of integrated information and to identify management gaps in the application of conservation efforts. We interconnect three key pieces of information: 1) the distribution of pressures, 2) information on species and habitats, divided into two databases: the spatial distribution and the sensitivities to pressures, and 3) the conservation measures. We generated and organized the information on conservation measures (sourced from the web-based database, see details below), and we also developed the pressure distribution maps, based on the pressures that are expected to impact the most species/habitats. In the case of species and habitat information, both their distribution and sensitivity to pressures were collected from the Report on the EU Directives developed in Catalonia (Generalitat de Catalunya, 2019). More details are explained in subsequent sections. We framed our work in the EU policy context, considering the potential value of our information for policy instruments and their implementation. By combining these pieces of information, we could, for example, explore questions such as evaluating hotspots of impacts (e.g., areas that concentrate co-occurrence of pressures and species/habitats sensitive to them) or identifying management gaps (e.g., the incidence of pressures on species and habitats that have not been covered with reported measures spatially and non-spatially, Fig. 2).

2.1. Database on conservation measures

To collect data on conservation measures, we defined conservation measures as any active action taken in the territory for biodiversity conservation purposes. Therefore, preventive and mitigation measures were not considered. We developed a web-based database where actors involved in biodiversity conservation in Catalonia (mainly governmental employees and managers of protected areas) could report information about active conservation measures carried out during the reporting period. The information requested consisted of a description of the measure implemented, the measure typology (according to the list provided by the EU EEA, 2020) the location of implementation (both inside or outside Natura, 2000 sites), the budget invested, which habitats and species are direct or potential beneficiaries, and the pressures that each measure particularly addressed (Appendix A).

For the spatial management gap analyses, we used the location of each measure that was provided in the database, i.e., in which Natura 2000 sites were implemented or whenever the measure was

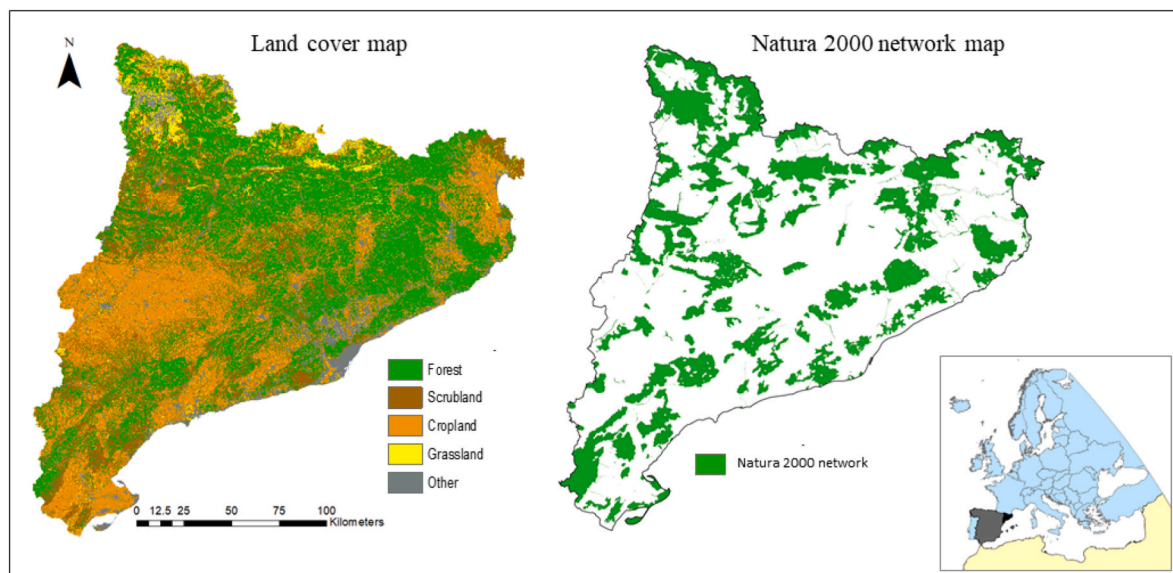


Fig. 1. The main land cover types in Catalonia are on the left. Natura 2000 network distribution (on the right) and Catalonia's location in Europe (on the downright corner). With very few exceptions, protected areas in the region are included in the Natura 2000 network. Sourced from: www.icgc.cat.

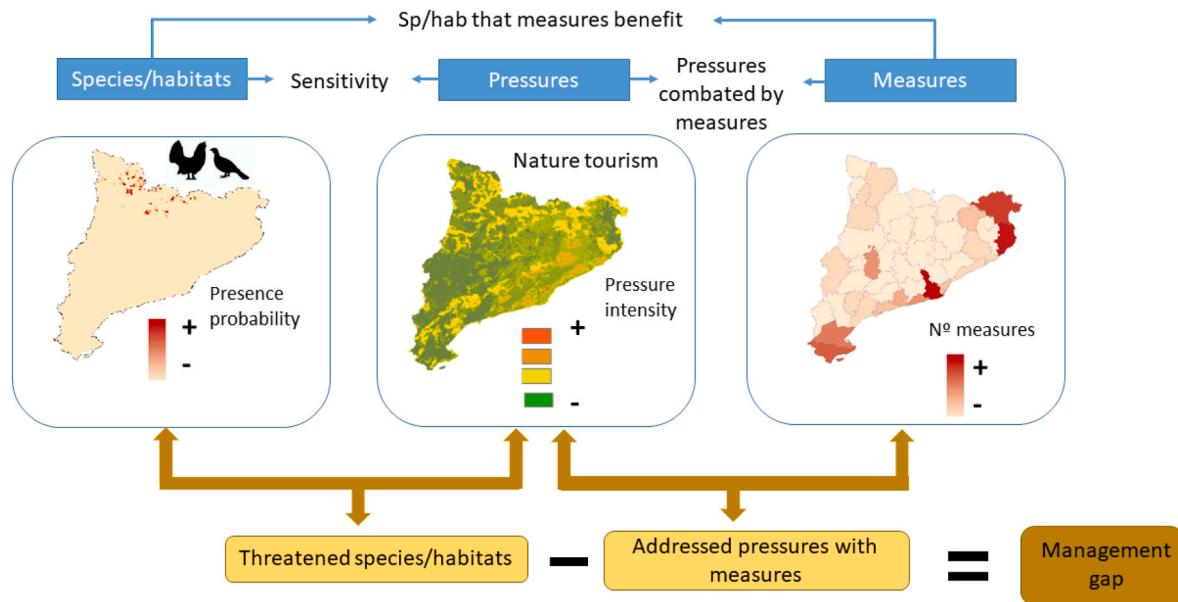


Fig. 2. A general framework of the database integration. The different combinations of the three pieces of information (species/habitats, pressures, and measures) allow us to identify the management gap: places where there are pressures, and species/habitats that are sensitive to the pressure (threatened sp/hab) minus the pressures addressed for sensitive sp/hab (addressed pressures by measures). The maps are examples of species distribution (western capercaillie, *Tetrao urogallus*), a pressure that the species is sensitive to (Nature tourism and outdoor sports), and the measures, in this case, the ones related to that pressure.

implemented outside the Natura 2000, in which county. If possible, we adjust the measure's location by choosing the overlapping surface between the measure and the distribution of the species/habitats that benefit from the measure. For instance, if a measure for the western capercaillie (*Tetrao urogallus*) was located in the Alt Pallars National Park (771,83 km² area) but the known distribution of the western capercaillie is restricted to only 60 km² inside the park (this is just an example) then, the location for the measure was reduced to that area. If there is more than one species or habitat, we join all the distributions. We translated the information of the measures implemented to the 1 km² grid cell.

2.2. Distribution of pressures

To create the pressure maps, we sorted the full list of pressures included in the measures database that we collected. These pressure typologies and definitions follow the European Commission pressures and threats list specified for the last Report on the EU Directives (EEA, 2020). For the spatial management gap, we selected the pressures that impacted the most species/habitats and that could be transferred to a map. We collected 48 pressures that we grouped in 8 pressure maps and that we used for subsequent analyses (See Appendix B to see the aggragation details). The pressures that we considered were: nature tourism and outdoor sports, agricultural intensification, water pollution, transversal water barriers, livestock overgrazing, land abandonment (assessed by two pressure maps: Abandonment of traditional uses and afforestation), and forestry management (Table 1). Final maps for each pressure were at 1 km grid cell resolution and are included in Appendix C. Further information on the followed methodology for each pressure is detailed in Appendix D.

2.3. Distribution and sensitivity of species and habitats

We considered for the analysis the species and habitats with available information that occur in Catalonia and that are included in Annex I of the Birds Directive or Annexes I or II of the Habitats Directive. We decided to focus on this group of species and habitats because there is a legal mandate at the EU level to protect and conserve them. We only

included species outside these annexes if they appeared in the measures database. We sourced information on the distribution of species across Catalonia at 1 km² resolution from the Report on the EU Directives in Catalonia for the 2013–2018 period (Generalitat de Catalunya, 2019), except in the case of the winter distribution of birds, that was not required in the reporting, and we sourced from the atlas of wintering birds of Catalonia at the same resolution (Herrando et al., 2011). We collected presence/absence distribution data for 175 species: 11 amphibians, 4 arthropods, 85 bird distributions (considering independently the winter and the breeding distribution if both occur in the territory –57 birds species–), 1 bryophyte, 8 fishes, 2 invertebrates, 1 lichen, 34 mammals, 18 reptiles, and 11 vascular plants. In the case of habitats, understanding habitat as it is defined in the Habitat Directive, we collected information from the Catalan Habitat Cartography (HIC v2), which can be expressed as the percentage of occupancy of habitat in each 1 km² grid cell. We translated habitat occupancy information into presence-absence for subsequent analyses, by considering that a habitat was present in a certain cell if the percentage of occupancy was $\geq 1\%$. We compiled information on the spatial distribution of 74 habitats of community interest under the EU Habitats Directive. Therefore, we finally had 249 habitats and species distributions at 1 km² grid cell resolution.

We also used a relational database with information on each species and habitat sensitivity to each pressure. Sensitivity to pressures was defined as the negative effect that a given pressure has on the species/habitat and was assessed species by species based on literature reviews and expert knowledge, including Red books and Conservation Plans of endangered species, the Manual of Cartography of Habitats of Community Interest (CHIC) and the database of EU Member States Report on Natura 2000 Directives. The sensitivity database does not include positive sensitivities but, each species/habitat could be sensitive to more than one pressure simultaneously. This database was sourced from an Interreg European project to support and inform environmental impact assessments and determine compensatory measures, where the sensitivity of the species/habitats to a pressure in a given location was used to determine the impact of a project that caused certain pressures where certain species/habitats occur (IAIA, 2019). This database was also used for the report on the Birds and Habitats Directives of Natura (2000)

Table 1
Pressure sources and brief Descriptions.

	Pressure name	The code OF EU pressures included	Description
1	Nature tourism and outdoor sports	F05, F06, F07, F09, F14, F24	Pressures that could appear due to the development of sport, tourism, and leisure activities outside the urban and recreational zones. Including the development of infrastructure, waste deposition, or treatment apart from the pressure that could cause the activities per se.
2	Agricultural intensification	A01 to A05, A15 to A17, A19, A21,A23, A25,A26,A34,A36	Pressures related to conversion to agricultural land or a more intensive agricultural, and all the pressures related to agricultural changes or activities. Such as changes in the landscape or soil composition, fertilizers, and water pollution among others.
3	Water pollution	F12, J01	Pressures related to sources of surface and ground freshwater contamination
4	Transversal freshwater barriers	A33, B27, D02,F33, G20,K03 to K05	Pressures related to the alteration of hydrological flow in freshwater bodies due to different elements: dams, water abstractions, or other barriers.
5	Livestock overgrazing	A09	Pressures related to intensive or overgrazing by livestock in agricultural and agroforestry habitats (e.g. pastures, meadows, pastured forests) where grazing causes damage to vegetation or soil.
6	Abandonment of traditional uses	A06, A07	Scrubland increase map: Pressure related to abandonment processes taking place in the territory that shifted the landscape from pasture, grassland, or other types of open areas to scrubland dominance landscape.
7	Afforestation	B01, B04	Forest increase map: Pressure related to abandonment processes taking place in the territory that shifted the landscape from pasture, grassland, or other types of open areas including scrubland to forest areas.
8	Forestry management	B02, B05,B06,B09, B11 to B13,B15,B16, B17,B21,B22	Pressures related to all kinds of forest management activities.

network where sensitivity is a key aspect for determining the conservation status and trends of species and habitats of EU Community Interest ([Generalitat de Catalunya, 2019](#)). Further details on this database are in [Appendix E](#). When we combined this information with the list of species and habitats considered, we found 986 species/habitat combinations that are sensitive to a given pressure. When combined with the spatial distribution of both species/habitats and pressures, resulted in 1,277,130 individual combinations of pressures x species/habitat across the territory.

3. Gap analysis

We divided the gap analysis into two phases, a non-spatial gap analysis, where we interconnected tabular information (see [Appendix A](#)), and a spatial gap analysis, where we take into account the spatial distribution of species/habitats, measures, and pressures.

2.4. Non-spatial gap analysis

We evaluated the coverage by the measures of species that were sensitive to individual or multiple pressures (comprising the identified 986 combinations), which specifically considered the interplay of measure, pressure, and species/habitat combinations. This involved merging two databases: one with information on pressure x species/habitats and the list of measures reported by stakeholders for the period 2013–2018. We relied on the information provided in the reported measures database, specifically, the information indicating which pressure each measure addressed (see [Fig. 2](#)). We assumed that the measure addressed the respective pressure.

We identified combinations of species/habitat x pressure lacking recorded management measures (what we called the management gap). This identification was crucial for calculating the non-spatial management gap, and we performed this analysis individually for each species/habitat. To report the results of this gap analysis, we grouped species and habitat information into 9 taxonomic groups, including 7 for vertebrates and invertebrates animals, 2 for plants, and 1 group for habitats. For each group, we calculated information about the total and the average number of measures and the allocation of the measures (inside, outside the Natura, 2000 network, or both). We also calculated the mean number of pressures associated with each measure per group.

2.5. Spatial gap analysis

To evaluate how well the 1,277,130 grid cell combinations of pressures x species/habitats had been covered with the measures reported, we developed a spatial analysis by using the information available on the distribution of species/habitats, pressures, and measures described above. In this case, to assess this gap, we used the same logic as the non-spatial gap analysis (i.e., we identified which pressure x species/habitat combinations had no management measures reported, but for each 1 km grid cell). We could only develop the spatial gap analysis for the species/habitats and pressures for which we had spatial data. We combined the spatial data on the distribution of species/habitats and the pressures they are sensitive to create a spatial layer of pressure incidence for each combination of pressure x species/habitat individually. We then combined that layer with the spatial distribution of measures to identify where pressures had been addressed. We repeated the same analysis for different pressure thresholds (50, 75, and 90 % of pressure intensity). The threshold determined whether the intensity of the pressure was considered high enough to impact the species/habitats in that grid. To report the results, we also determined, for each group of species and habitats, the overall coverage (number of 1 km grid cells) where pressure x species/habitat combination was addressed by measures.

3. Results

3.1. Non-spatial gap analysis

We compiled information for a total of 691 measures in our web-based database, covering 162 pressures for 239 species and 91 different habitats, from 61 different users of 31 institutions acting from the local (e.g. a national park) to the sub-national level (Catalonia). There were 54 different types of measures, and their distribution among the potentially beneficiary groups of species and habitats was not homogeneous among them. There were important differences between and within taxonomic groups ([Table 2](#)). Plants and mollusks had the lowest number of measures standing as the groups with the biggest management gap. Birds and habitats were the better-covered groups regarding the total number of measures. Amphibians received the highest average number of measures per species, while plants had the fewest. Mammals stood out as the group with the highest number of pressures addressed per measure (18.9 ± 88.4). We detected high standard deviations within groups, indicating that there were species with considerably more

Table 2

Distribution of conservation measures collected in the measures database among groups. Dark colors highlight the highest number in that column. N2K stands for Natura 2000 network.

Group	Σ Measures	Mean measures/ specie	Σ Measures inside N2K	Mean measures inside N2K	Mean number of pressures per measure
Amphibians (N=10)	130	13 $\pm$ 6.9	99	6.6 $\pm$ 5.5	10.6 $\pm$ 21.8
Arthropods (N=12) and other invertebrates (N=2)	86	6.1 $\pm$ 7.9	60	5.5 $\pm$ 6.6	9.4 $\pm$ 44.8
Birds (N=150)	1165	7.9 $\pm$ 8.8	880	5.4 $\pm$ 6.6	12.6 $\pm$ 43.9
Fishes (N=12)	80	6.7 $\pm$ 6.7	47	5.9 $\pm$ 6.1	3.2 $\pm$ 9.1
Mammals (N=36)	291	8.1 $\pm$ 5.4	183	5.1 $\pm$ 3.7	18.9 $\pm$ 88.4
Mollusks (N=4)	22	5.5 $\pm$ 3.0	15	3.0 $\pm$ 1.9	2.9 $\pm$ 2.7
Reptiles (N=9)	91	10.1 $\pm$ 14.6	62	8.9 $\pm$ 12.7	3.7 $\pm$ 12.8
Vascular plants (N=14) and non-vascular plants (N=2)	42	2.6 $\pm$ 1.8	15	1.7 $\pm$ 0.9	5.2 $\pm$ 13.9
Habitats (N=91)	981	10.8 $\pm$ 12.2	740	8.1 $\pm$ 9.8	8.2 $\pm$ 55.5
TOTAL	2888	6.8 $\pm$ 3.7	2101	5.0 $\pm$ 2.7	8.7 $\pm$ 31.1

measures reported than others. For instance, forest habitats had more than double measures than scrubland habitats, and the Bonelli's eagle (*Aquila fasciata*), recorded 61 measures, when the mean for birds was 7.9 measures per species. For reptiles, the high standard deviation was driven by two species European pond turtle (*Emys orbicularis*) and Iberian Pond turtle (*Mauremys leprosa*) which concentrated more than 2/3 of all measures per reptile, 40 and 29 respectively.

3.2. Spatial gap analysis

Of those 239 species and 91 habitats recorded in the measures database, we could only make the spatial gap analysis for 175 species and 73 habitats that we had spatial distribution. The highest concentration of species/habitat x pressure co-occurrences was located in the northeast of Catalonia, occurring in and around Montseny Natural Park and the Garrotxa Volcanic Zone Natural Park and the headwaters of the Ter river, both included in the Natura 2000 network. These areas had over 75 pressure x species/habitat combinations per 1 km² grid cell (Fig. 3, A, upper map). After filtering out those combinations that had been addressed through measures (Fig. 3, bottom map), we found spatial differences in the management gap. Whereas we found some areas where multiple measures had been implemented covering partly the existing pressures (at least for some species), in some other areas the management gap was large due to the lower number of measures reported. For example, in the northeastern area that we highlight above as a hotspot for pressure x species/habitats combinations, while some pressure has been addressed with measures, it was still the area with the larger spatial management gap (orange colors in Fig. 3, caption A). However, there were areas where pressures had been well-addressed (purple-to-brown colors in Fig. 3 bottom map), such as the Central Depression of Catalonia (central west part of the region, Fig. 3, B), which is a flat area traditionally devoted to agriculture. Even though, looking into which pressures were addressed in this last area, they were pressures related to measures in riparian forests and other non-agricultural areas. Also, the Ebro delta region (Fig. 3, caption C), which is a large wetland with a high concentration of biodiversity that coexists with agricultural practices such as rice cultivation, showed a large proportion of pressures addressed with measures. Other areas in the southwest were highly addressed as well (Fig. 3, right part of caption C).

A total of 304,598 measure x pressure x species/habitat combinations were found, this implied that 24% of all the combinations of pressure x species/habitat were addressed by measures. Although, there were important differences across taxonomic groups and habitats. Despite plants having very few measures recorded in the database on average (Table 2), they were the best-covered group (there were measures for 50% of all the cells with species x pressure co-occurrences; Fig. 4). Also, they have the smallest surface on average compared to the other groups. However, this number could be mainly driven by a single species (*Spiranthes aestivalis*; not included in Annex II of the Habitats Directive) for which measures had been reported in a large area

within its distribution range. The taxonomic groups with the lowest number of measure x pressure x species/habitat combinations were birds, fishes, and habitats. Regarding the proportion of addressed pressures x species/habitats combinations included in the Directives, we could see that invertebrates had the higher proportion addressed compared to the total number of species in the group. On the contrary, amphibians and plants had a smaller amount. Habitats are all included because all habitats were included in the Annex I of the Habitat Directive. When constraining the gap analyses to the areas with the highest pressure intensities (90%), we found that measures were generally implemented where pressures were more intense (Fig. 4). Invertebrates were the only group where measures were not targeted in the more intensive pressure areas, as more percentage was addressed when considering 75% of pressure intensity.

4. Discussion

Using the example of Catalonia, we showed that the strategic integration of already existing information enhances the wide potential of the data to inform decision-making and conservation action. Organizing and making better use of the information is sometimes a bigger obstacle than the lack of information itself (Buxton et al., 2021). Here we exemplified how to identify management gaps in the territory by integrating spatially explicit information on pressures, species, habitats, and measures. We focused on the collection of conservation measures already developed in the territory. We found that data on conservation measures was not being collected effectively in the territory so far and that the conservation measures that we gathered were not balanced between taxonomic groups and space. We identified management gaps for some taxa (e.g., vascular plants and mollusks), and areas in Catalonia such as the northeast of the region that had the highest spatial management gap (i.e., pressure x species/habitat combinations with no measure reported). The improvement of the information incorporated in the measures database and the application of a framework of data integration like ours could give a broad vision of the conservation needs in the territory through the application of a simple framework.

Our results show that there were important biases in the conservation measures across and within taxa, especially towards birds, which received more amount of conservation measures than average but have differences between bird species as well (Table 2). There was a remarkable lack of measures for groups such as invertebrates (arthropods and mollusks) and plants. These results supported recent studies on the distribution of funding across the EU, that highlight that animal species received three times more funding than plants on the LIFE programs (Adamo et al., 2022) and that investment in vertebrates is six times higher than for invertebrates (Mammola et al., 2020). Hence, biases toward birds were not unexpected given that they have their own Bird Directive, and they are a group historically used as indicators of biodiversity value (Fraixedas et al., 2020). This is reasonable as we tend to safeguard what we know best, and therefore conservation efforts

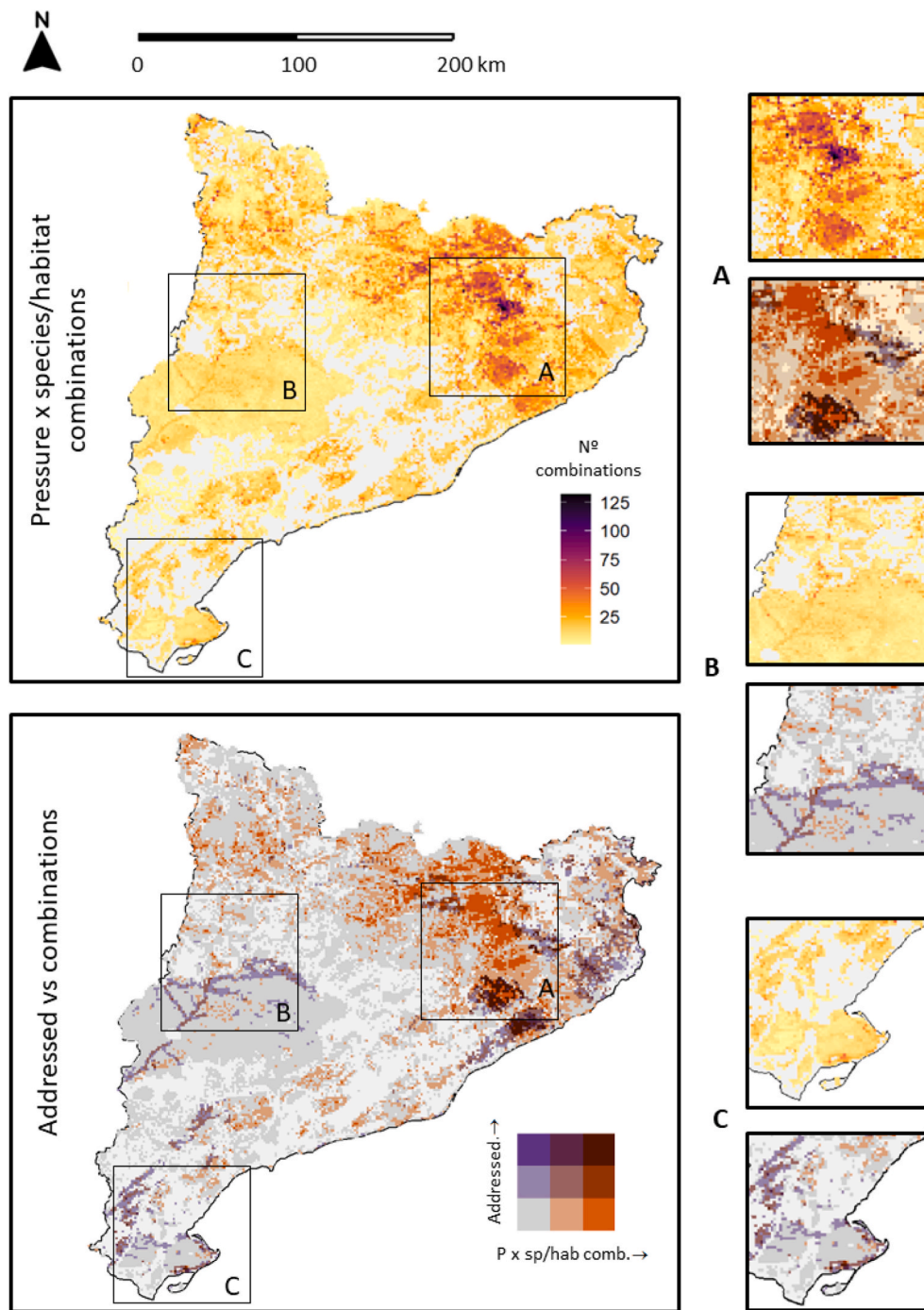


Fig. 3. Pressure x species/habitat combinations (map above) and management gap (map below). The map below is a bivariate map that shows the places where relations were addressed (increasing dark purple color) versus the places with a higher number of combinations (increasing dark orange-brown color). The pressure threshold used was 75%.

could be biased toward those species/habitats or sites. In this way, “stereotype” species, charismatic, or aesthetic values are also reasons that influence mismatches between taxonomic groups’ efforts (Britnell et al., 2021). Also, during the web-based compilation process for the conservation measures, we were aware that there could be a bias in the gathering of measures among taxonomic groups as well as in the spatial distribution of them, and that this could be affecting the results obtained.

There is consensus about the lack of adequate distribution of conservation efforts. Non-threatened species receiving more conservation attention than threatened ones (Hermoso et al., 2017) or unbalances

between investment and biodiversity values among European countries (Sánchez-Fernández et al., 2018) are some of the described issues. In our case, the distribution of effort across threatened and non-threatened species was influenced in part by the use of the Birds and Habitats Directives lists as a proxy of protected species/habitats. These lists have biases towards Central European species, so considering regional (Catalan or Spanish) legal framework could provide different results. At the same time, a significant portion of well-known species and communities in Catalonia is concentrated in the northeast of the region, reflecting the geographic distribution of the pressures they face. Standardizing species/habitat knowledge in the field of conservation science

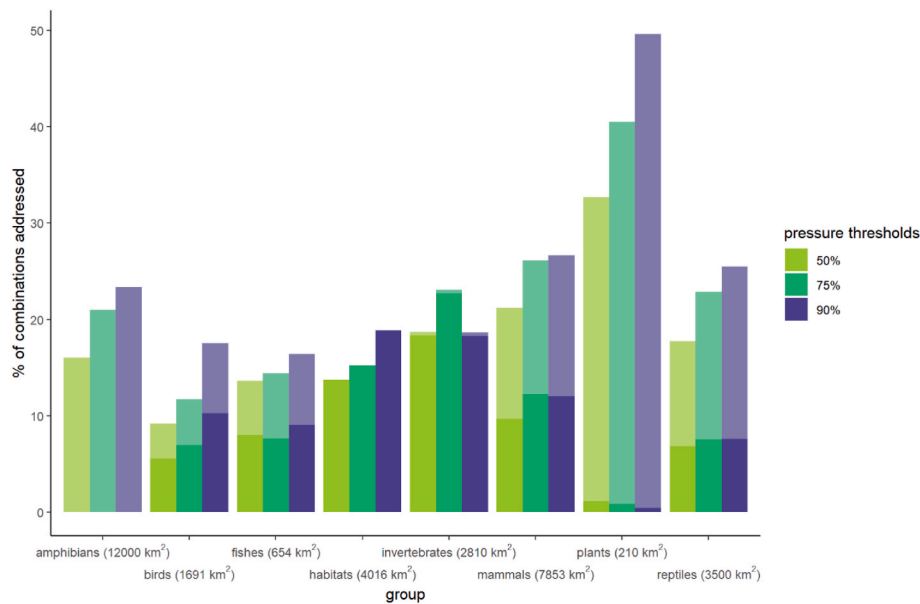


Fig. 4. Percentage of pressure x species/habitat combinations addressed (i.e., with some measure reported) per group. The different bar colors represent different values of pressure intensity. The darker color inside each bar indicates the number of combinations addressed corresponding to species/habitats included in the Habitat or Birds Directives. Km² represented the mean area occupied by each group in Catalonia (Total surface 32,000 km²).

is challenging, given that our comprehension of each species influences the allocation of conservation efforts, potentially resulting in imbalances (Darwall et al., 2011). Furthermore, the spatial management gap in the northeast coincides with important protected areas in the region, which may indicate that biodiversity and pressures in protected areas have been more studied compared to other unprotected areas (García-Macía et al., 2021). The identification of management gaps also allows the identification of knowledge gaps concerning species and habitats that have been traditionally unstudied.

The explicit consideration of the spatial dimension in our analysis was especially insightful. The analysis of the spatial management gaps provides insight into areas where resources are either inadequately recorded or insufficient to address the pressures faced by species and habitats. This gap analysis not only identifies regions requiring more attention but also highlights where existing measures effectively address pressures, such as the south of the region. This information is valuable, allowing for the assessment of the effectiveness of measures and informing future actions. Additionally, we detected that an elevated number of measures reported for a species did not imply better coverage (i.e., more species/habitat pressure addressed by measures). For example, despite the reduced number of measures, pressures for plants seemed to be the most adequately covered by measures since they addressed the highest percentage of pressure x species/habitat combinations (Fig. 4). Birds and habitats, on the contrary, have the highest number of measures associated, but their pressures were less covered by the measures reported. These, however, might be a result of the higher average distribution ranges of birds and habitats (1691 and 4016 km² on average), which made it difficult for the measures to cover all sites where pressure x bird/habitat combinations occur. In addition, there is evidence of the higher cost of mammals and birds' conservation actions over other taxons, which could be driving these inconsistencies (Gordon et al., 2020). We also show that, generally, measures were developed in places where pressures were more intense. However, the debate over where it is more efficient to intervene against pressures is present in the scientific community (Butt et al., 2020). Pressures to species/habitats are easier to identify when the intensity of the pressure is high, which could lead to a decrease in the effectiveness of the measures by acting when the status of the species or habitats is already too compromised or causing an increase in the necessary effort invested (Tulloch et al.,

2015). Studies prioritizing pressure management give practical recommendations to guide how to address management gaps in a cost-efficient way (Carwardine et al., 2012; Hermoso et al., 2022).

Integration of conservation measures implemented in the territory with other sources of information is a crucial step to create mechanisms to translate knowledge into action and so, accomplish global conservation goals. The European Commission already has tools to integrate biodiversity knowledge with policy instruments, the Report on EU Directives, and tools to prioritize efforts in the future, such as the Prioritized Action Frameworks (PAFs). PAFs are multi-annual strategic planning frameworks that aim to provide a comprehensive overview of the measures needed to implement the Natura 2000, specifying the funding needs for these measures, and linking them to the relevant EU funding programs (EC, 2022). PAFs are aimed at planning measures not only inside but also outside the Natura 2000 network (mainly for the green infrastructures that contribute to the ecological coherence of the Natura, 2000 network). Our work could help plan management needs through the identification of management gaps where is a priority to act, and hence, helping the formulation of the new PAFs. Monitoring the effectiveness of the measures implemented is the following step needed to ensure that the pressures have been addressed with the measures (Navarro et al., 2017). The Report on EU Directives assesses the conservation status of species and habitats, allowing to find general patterns of the status of the populations (Generalitat de Catalunya, 2019; EEA, 2020), but does not monitor the effectiveness of the measures taken, which makes difficult to find efficient solutions. Integrating the development and results of the Report on EU Directives, with PAFs is a key step to make concrete recommendations to improve conservation action at smaller scales, such as regionally or individual Natura 2000 sites. PAFs have also been updated in recent years to better represent conservation priorities and to adopt a more regional management strategy. The approach that we developed here, has the potential for broader application in other regions if data is available. Its applicability is especially relevant for Member States of the EU, considering the shared policy requirements and temporal resolution of data across countries. However, challenges could also arise, such as issues related to data quality or mismatches in spatial resolution. The European Commission places a primary emphasis on achieving homogeneity in data collection and subsequent integration within the context of the Report on EU

Directives. Given that database integration could play an essential role in prioritizing action, it is important to ensure that the databases adequately cover not only pressures, as traditionally done (Allan et al., 2019a,b; Bowler et al., 2020) but also conservation measures.

Systematically collecting data, and the identification of shortfalls in biodiversity knowledge and its consequences is key for addressing global biodiversity problems. This includes addressing the lack of data in certain regions or species, or the quality of the available information (Hortal et al., 2015; Sutherland et al., 2011). In this way, we are aware that the resolution of the conservation measures information that we had is limited and that this could be biasing our results by underestimating the management gap. Ensuring data integrity poses a big challenge for the scientific community, given the intensified focus on digitalization and open access policies on biodiversity data over the last four decades. Data quality could have impacts on ecological analysis and interpretation (Augustine et al., 2024). Applying approaches such as the “fitness for use” (Veiga et al., 2017) or focusing on data quality strategies (Chapman, 2005), are key aspects for enhancing the available information. Consequently, improvements in data quality contribute to more accurate results in integration exercises, such as the one undertaken in our study. Also, we had in our measures database a lack of representation of some actors and stakeholders that actively work on the territory for biodiversity conservation, such as conservationist NGOs or land stewardship organizations. Engaging with more conservation actors and addressing their necessities so they align with ours, would enhance the data collection process (Buxton et al., 2021). At the same time due to methodological limitations, we were not including preventive measures or mitigation and compensatory measures. These conservation efforts might have led to pressure decreasing or species status improvements, our analysis would benefit from these considerations in the future.

Conservation success would not be possible if we do not focus on investing efforts efficiently from local to global levels (Díaz et al., 2020). At the same time, limited financial resources and implementation capacity make necessary the valuation of management action information jointly with the prioritization of conservation action (Auerbach et al., 2014; Williams et al., 2020). Interconnecting disciplines such as biodiversity, macroecological analysis, and prioritization tools to explicitly develop comprehensive knowledge frameworks is key to avoiding biases and improving conservation action success (Santini et al., 2021; Zárate et al., 2020). Analysis like ours could provide information to complete pressures and threats analysis by incorporating actual conservation efforts, that are normally included passively through, for example, the consideration of protected areas as places where biodiversity is better managed (Allan et al., 2019b). Our approach combined with monitoring and prioritization of future conservation efforts could guide decision-makers to focus action where is more needed. This approach not only assists in meeting policy requirements, such as the Natura 2000 network assessment and prioritized action frameworks (PAFs) but also contributes to the development of species-specific plans or natural parks reports.

CRediT authorship contribution statement

Mónica Lanzas: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Nuria Pou:** Conceptualization, Data curation, Supervision, Validation, Visualization, Writing – review & editing. **Gerard Bota:** Conceptualization, Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing. **Magda Pla:** Data curation, Validation, Writing – review & editing. **Dani Villero:** Data curation, Validation, Writing – review & editing. **Lluís Brotons:** Funding acquisition, Investigation, Supervision, Validation, Writing – review & editing. **Pau Sainz de la Maza:** Conceptualization, Project administration, Supervision, Validation. **Joaquim Bach:** Conceptualization, Project administration, Supervision, Validation. **Sara Pont:** Conceptualization, Project administration, Supervision, Validation. **Marc Anton:**

Conceptualization, Data curation, Writing – review & editing. **Sergi Herrando:** Conceptualization, Data curation, Writing – review & editing. **Virgilio Hermoso:** Conceptualization, Data curation, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data (Appendix A-E)

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2024.120247>.

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